Checkpoint 2 – Midterm (Weeks 5–8)

Focus: Prompt Architecture and Vector Indexing

1. Prompt Architecture

- **System Prompt**: *"You are a fitness trainer AI. Generate workout plans using the provided exercises."*
- **User Prompts**: Capture goals, duration, equipment, and fitness level.
- **Context Prompts**: Feed retrieved exercises into OpenAI so responses are grounded in the database.
- **Output Format**: Structured plans with days, sets, reps, and progression.

2. Vector Indexing

- **Implementation**: Use FAISS or similar for efficient search.
- **Indexing Strategy**: Organize embeddings by category (strength, cardio, flexibility).
- **Optimization**: Top-k search returns the most relevant exercises.
- **Example**: User says *"I want a 3-day strength plan"* → vector search retrieves Bench Press, Deadlift, Squats → chatbot generates a natural workout plan.